AVCDL Traceability

Revision

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Abstract

This document provides a visual overview of how the **AVCDL** achieves the traceability needed when creating a safety-critical, cyber-physical system.

License

Overview

The creating a safety-critical, cyber-physical system is a highly complex endeavor. An important aspect of such systems is the ability to ensure that every activity is supported by those preceding it in a manner which allows for tracing each decision back to the stated goals of the system. The AVCDL was constructed with this need in mind.

AVCDL Product Dependencies

The following diagrams shows the dependencies between the products from the AVCDL process requirements. These are presented as a dependency graph (reversed) using ISO standard flowchart symbols.

For more information regarding the interpretation of these diagrams, refer to the ***Understanding the Phase Product Dependencies Graph*** secondary document.

The following diagram shows the end-to-end AVCDL phase product dependencies.

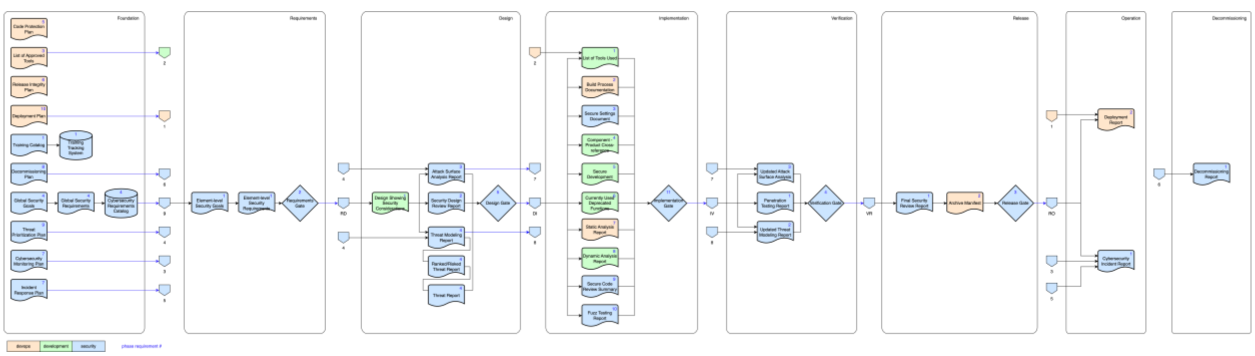

Figure 1 - AVCDL phase dependencies

The following pages provide a more document friendly view for each of the AVCDL phases.

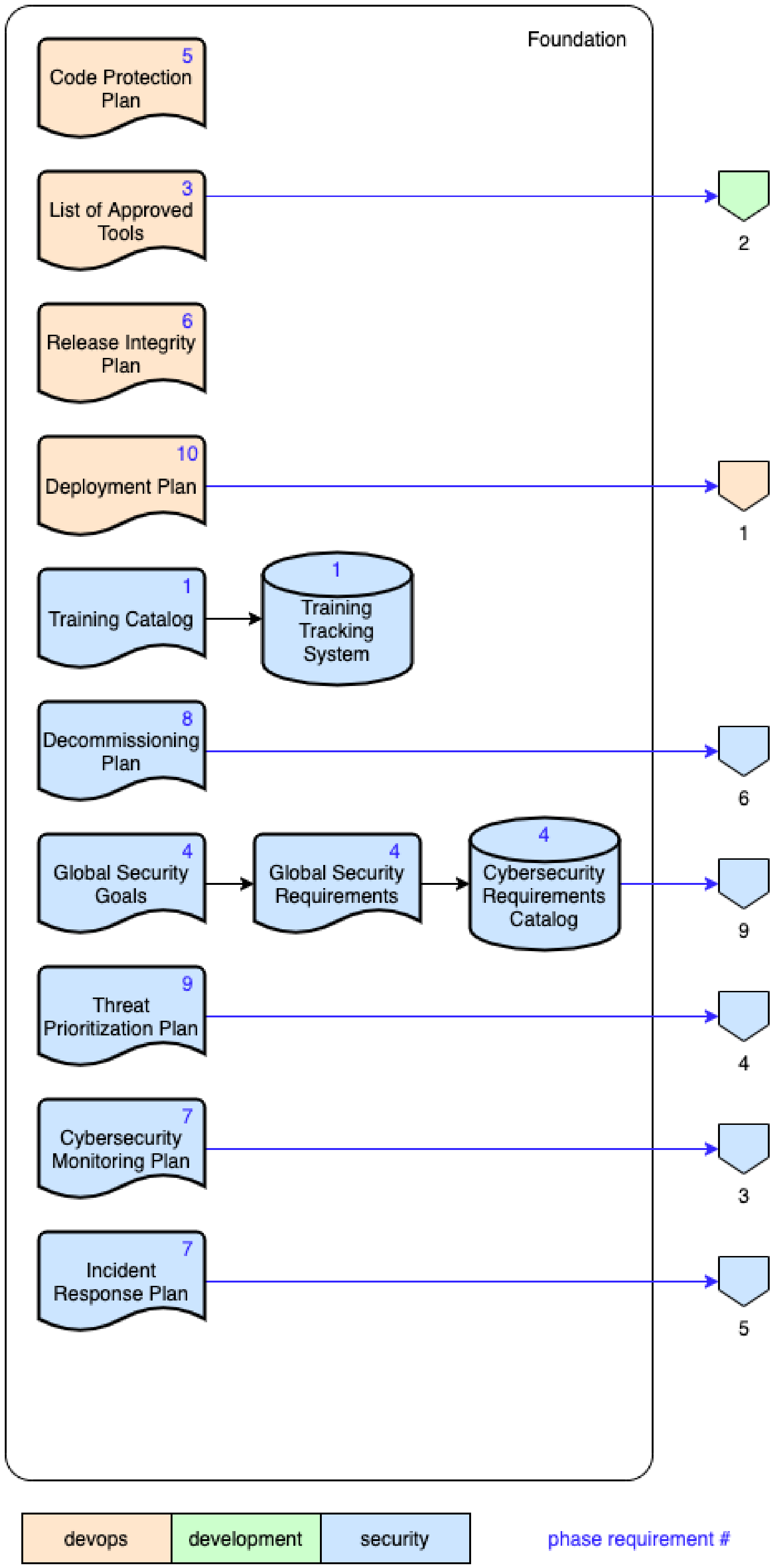

Figure 2 - AVCDL product dependencies – foundation phase

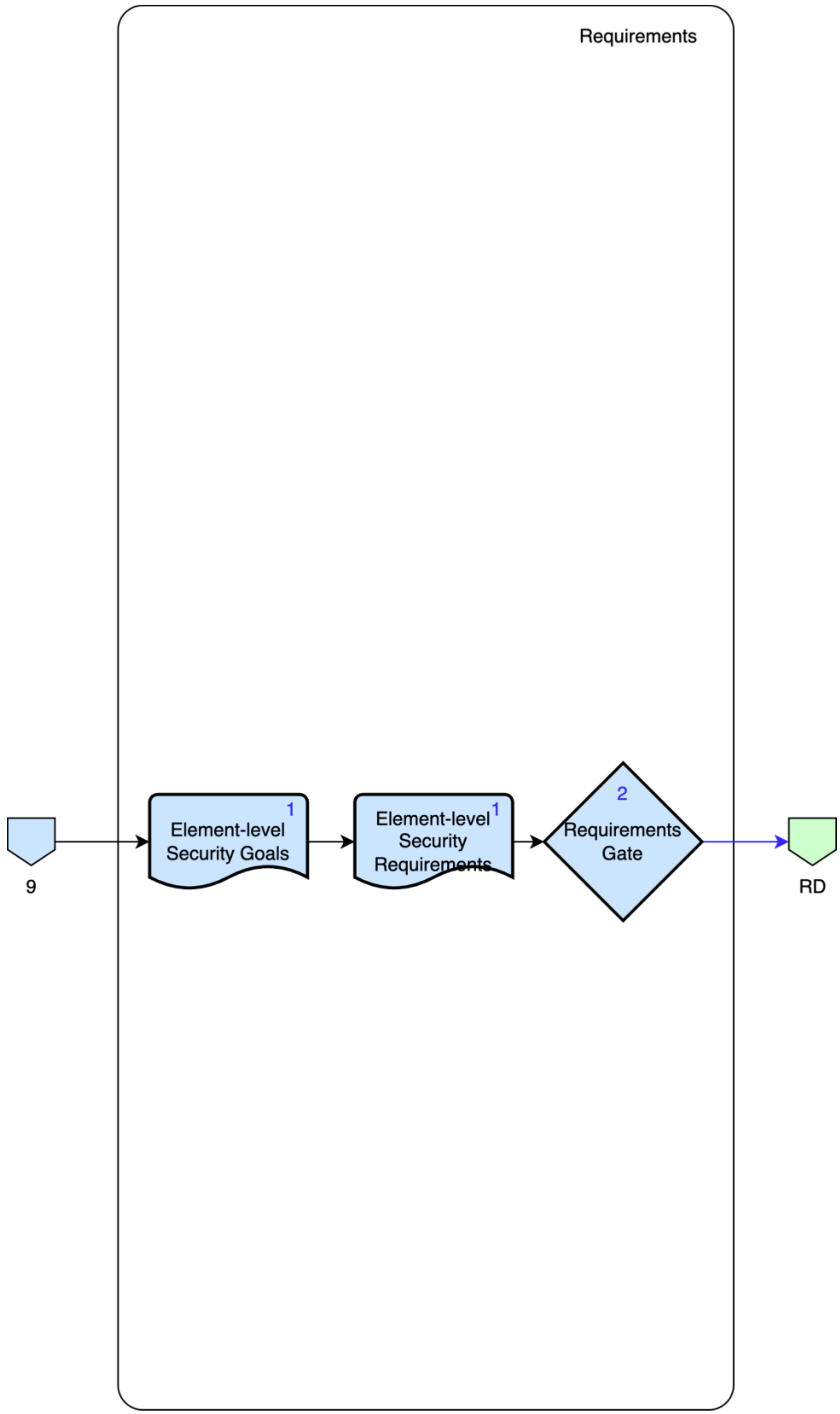

Figure 3 - AVCDL product dependencies – requirements phase

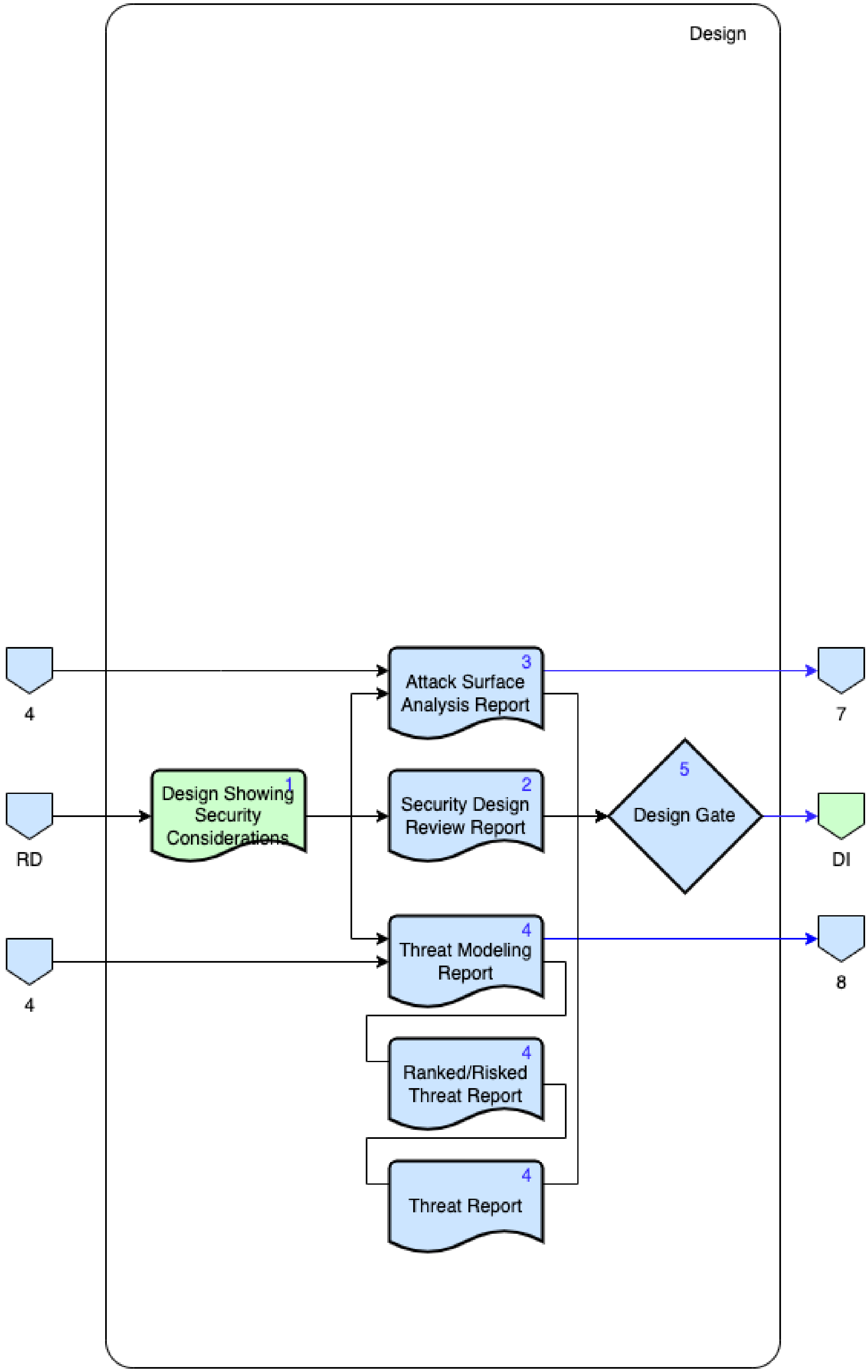

Figure 4 - AVCDL product dependencies – design phase

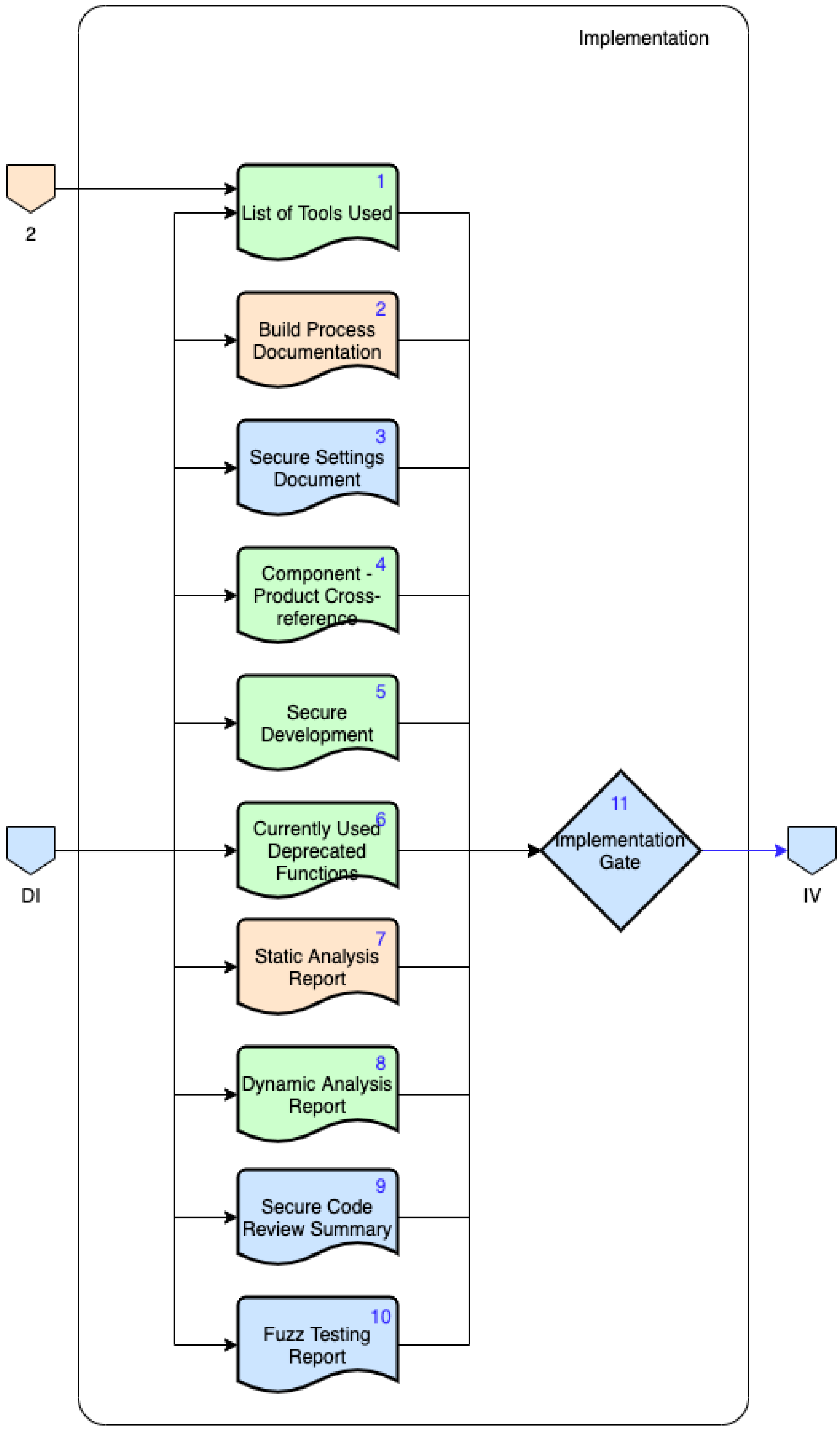

Figure 5 - AVCDL product dependencies – implementation phase

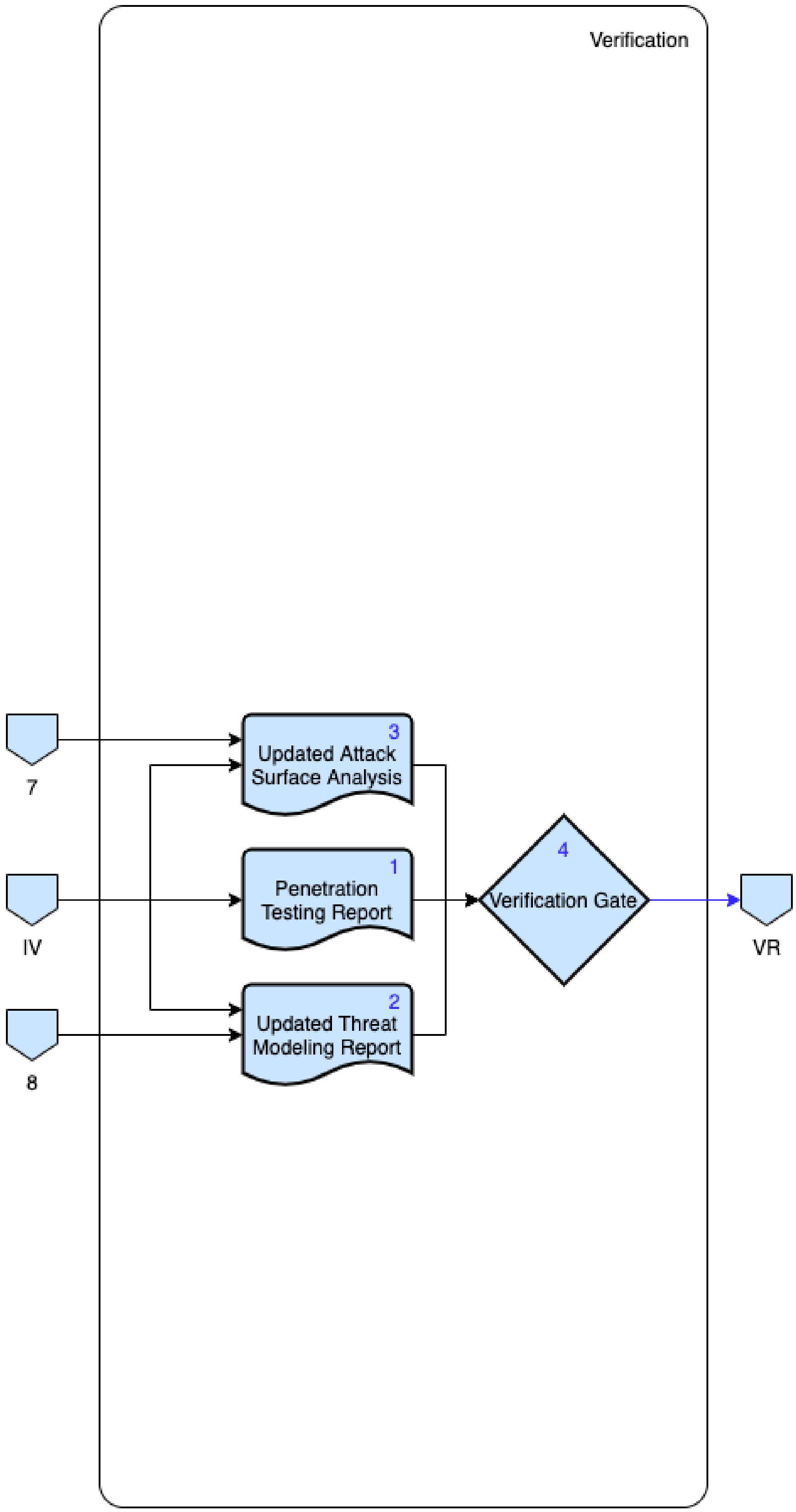

Figure 6 - AVCDL product dependencies – verification phase

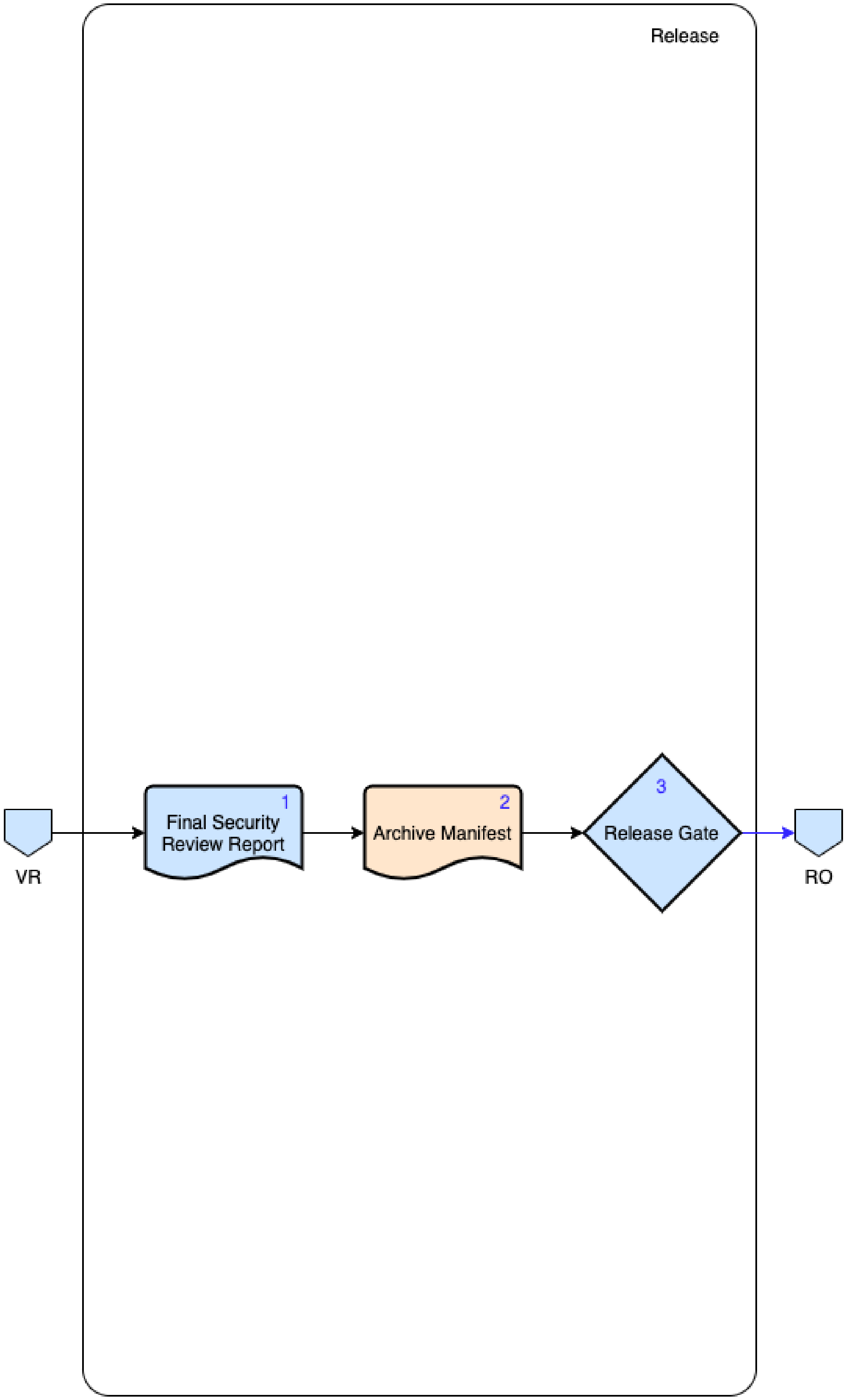

Figure 7 - AVCDL product dependencies – release phase

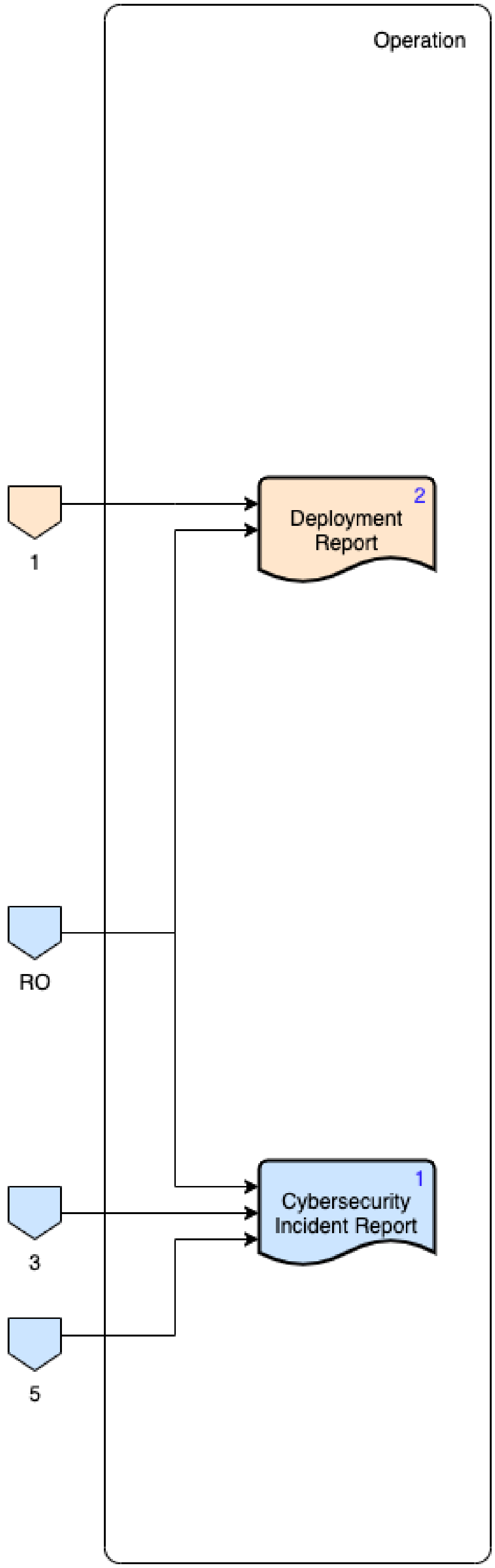

Figure 8 - AVCDL product dependencies – operation phase

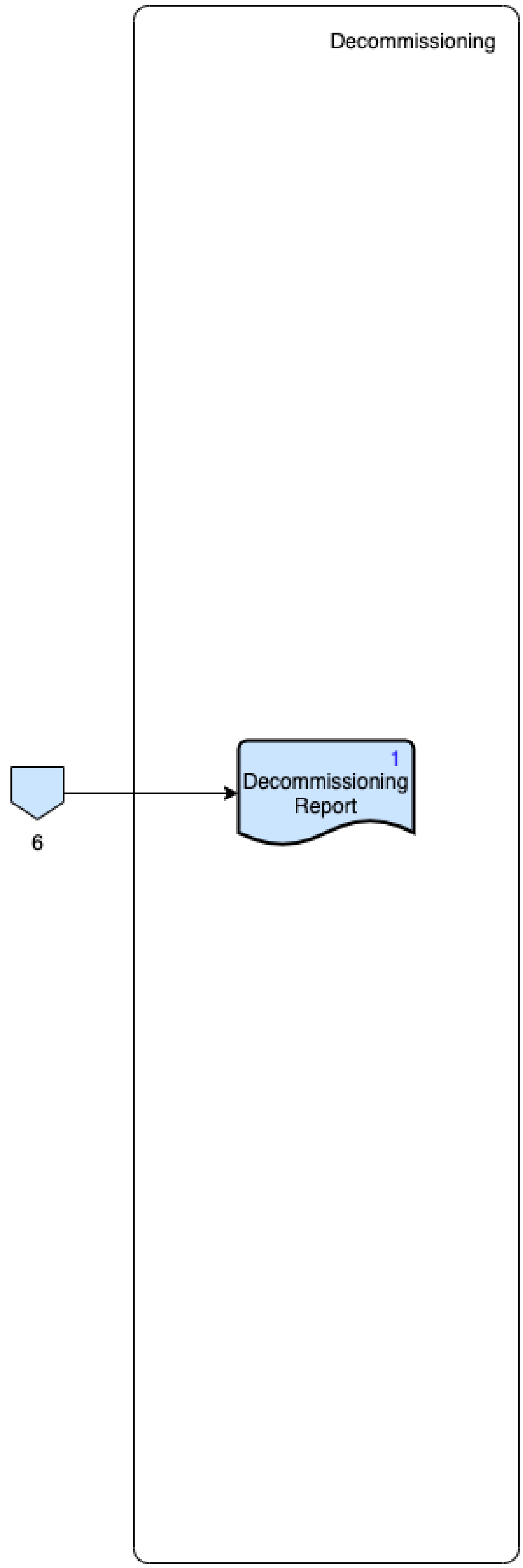

Figure 9 - AVCDL product dependencies – decommissioning phase